CHAPTER 11:
OPERATIONAL PLAN

LEARNING OUTCOMES

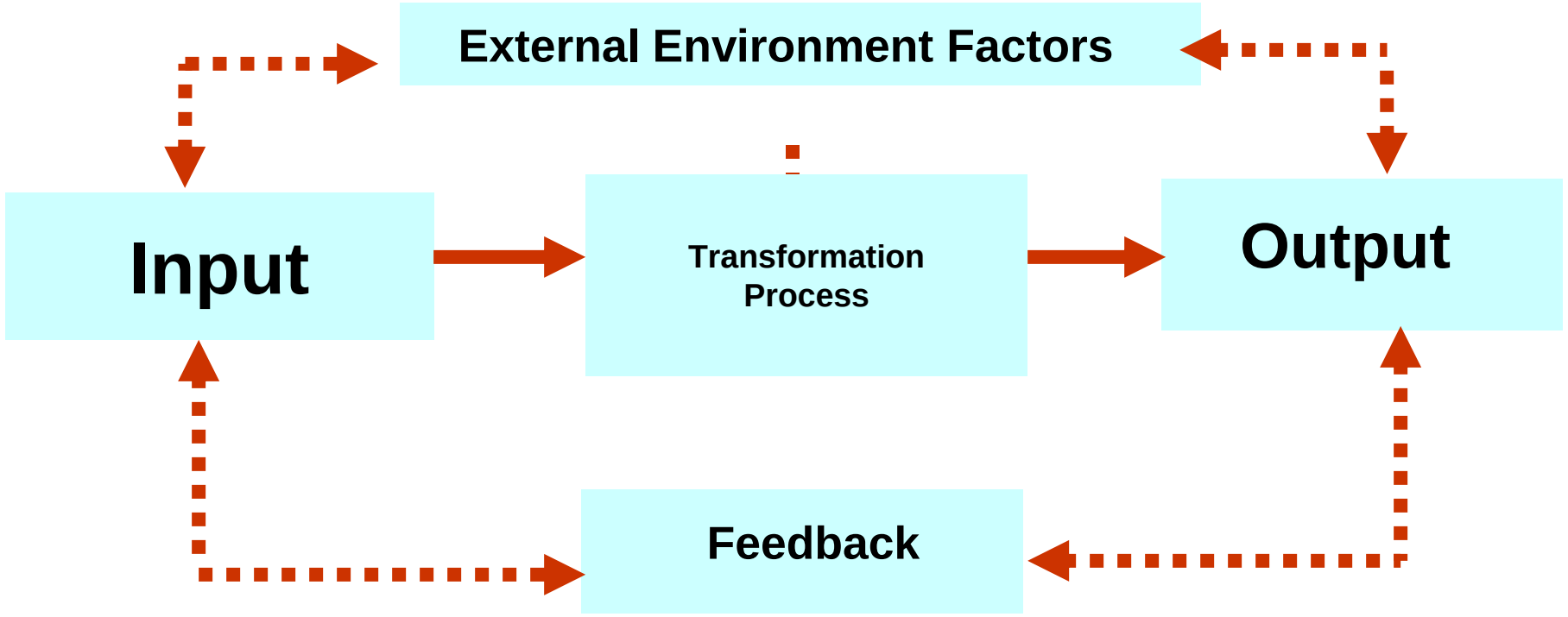
1. Understand the concept of operation in business
2. Understand the operations plan involving resource allocation to budget development, and decision making.
3. Prepare an operational plan

INTRODUCTION

- One of the important functions in a business organization.
- Process of marshaling business input to transform them into output in the form of products or services.
- Managing operations is crucial to make sure that the business is able to produce products or deliver services according to the specification, quality and time specified by the customer.

Operation Management Components

- Systematic direction and control of a process that transforms inputs into finished products or services.



Operational Plan

- One of the sections included in the business plan.
- Demonstrate the operational aspect of the business that is crucial in transforming the business input into output.
- Managed efficiently in order for the products or services to be delivered to clients according to their specifications, the required quantity and schedule demanded, and of minimum cost entailed

Why Operational Plan?

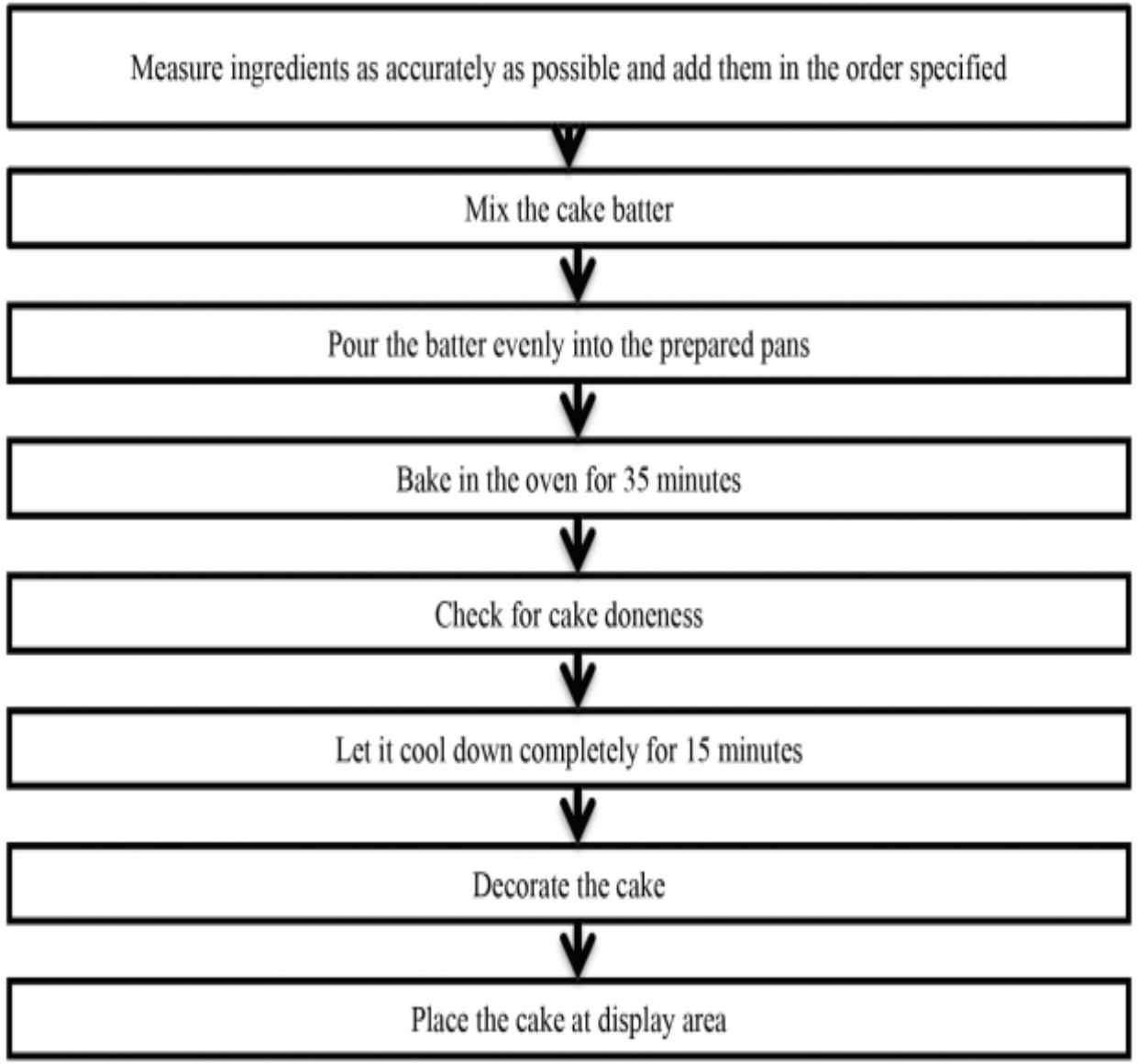
- Determine company performance & achieve objectives.
- Ensure the business is able to produce the product or to provide the services.
- Ensure company makes maximum profit with lowest possible cost.

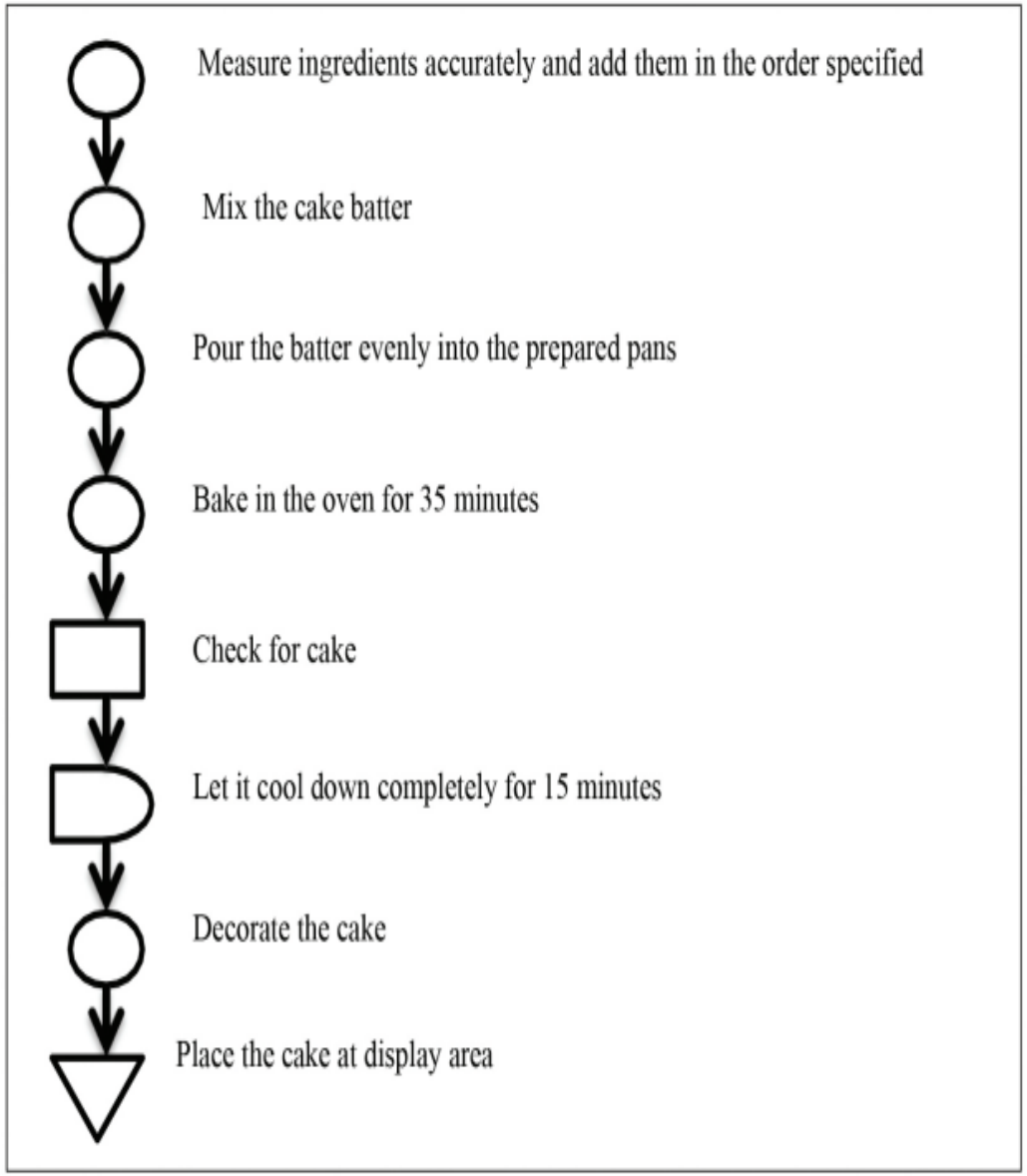
OPERATION PLAN

1. PROCESS PLANNING
2. OPERATION LAYOUT
3. BUSINESS & OPERATION HOURS
4. LICENSE, PERMIT & REGULATIONS
5. PRODUCTION PLANNING
6. MANPOWER PLANNING
7. MACHINE AND EQUIPMENT PLANNING
8. MATERIAL PLANNING
9. OPERATION BUDGET

1. Process Planning

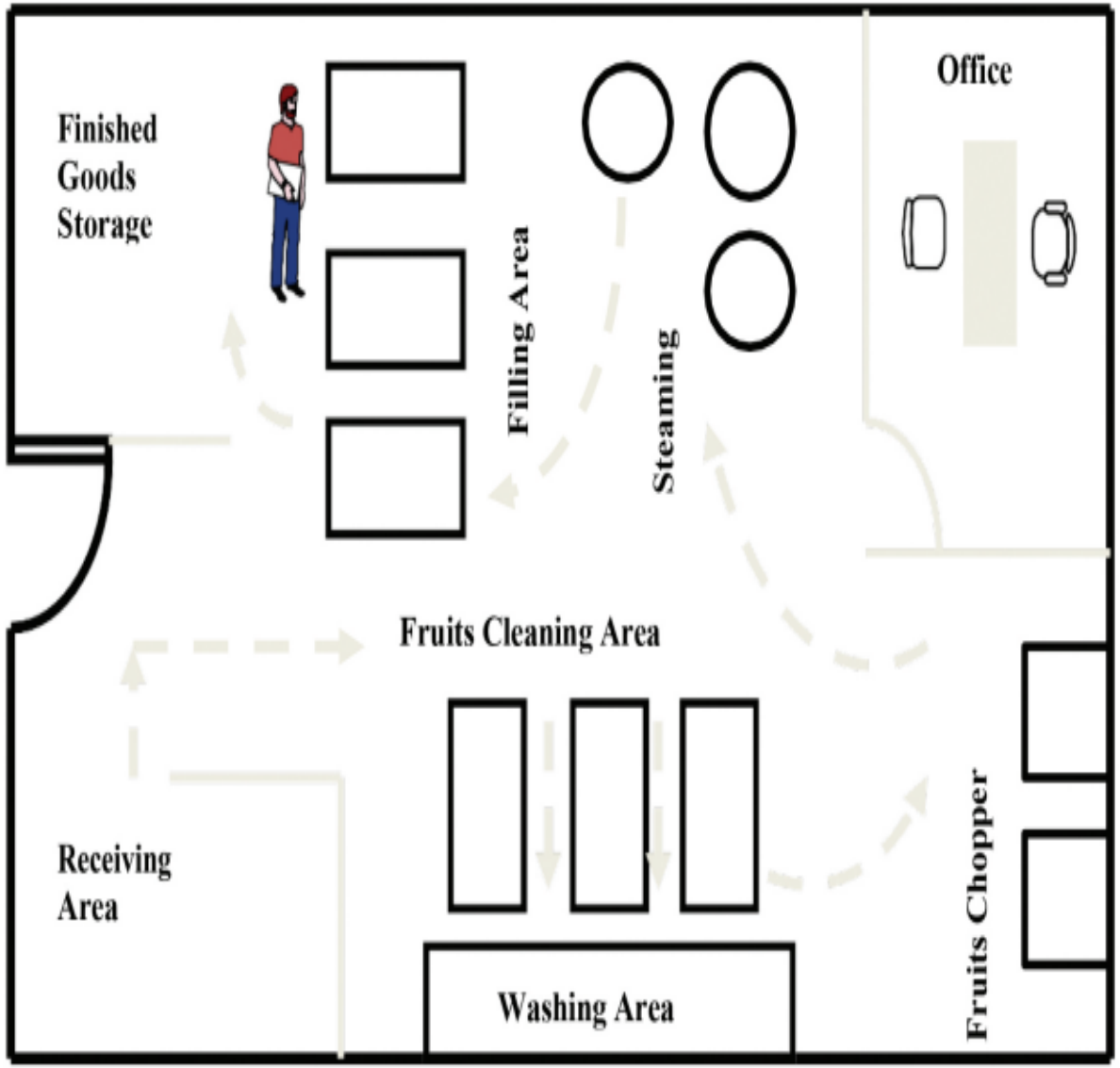
- Step- by-step process from selecting resources to the creation of a finished output.
- Establishing a proper sequence of activities is necessary to ensure the whole operation
- Aspects of creating the finished product are being done in an efficient and proper manner.
- Use flow chart or a process chart

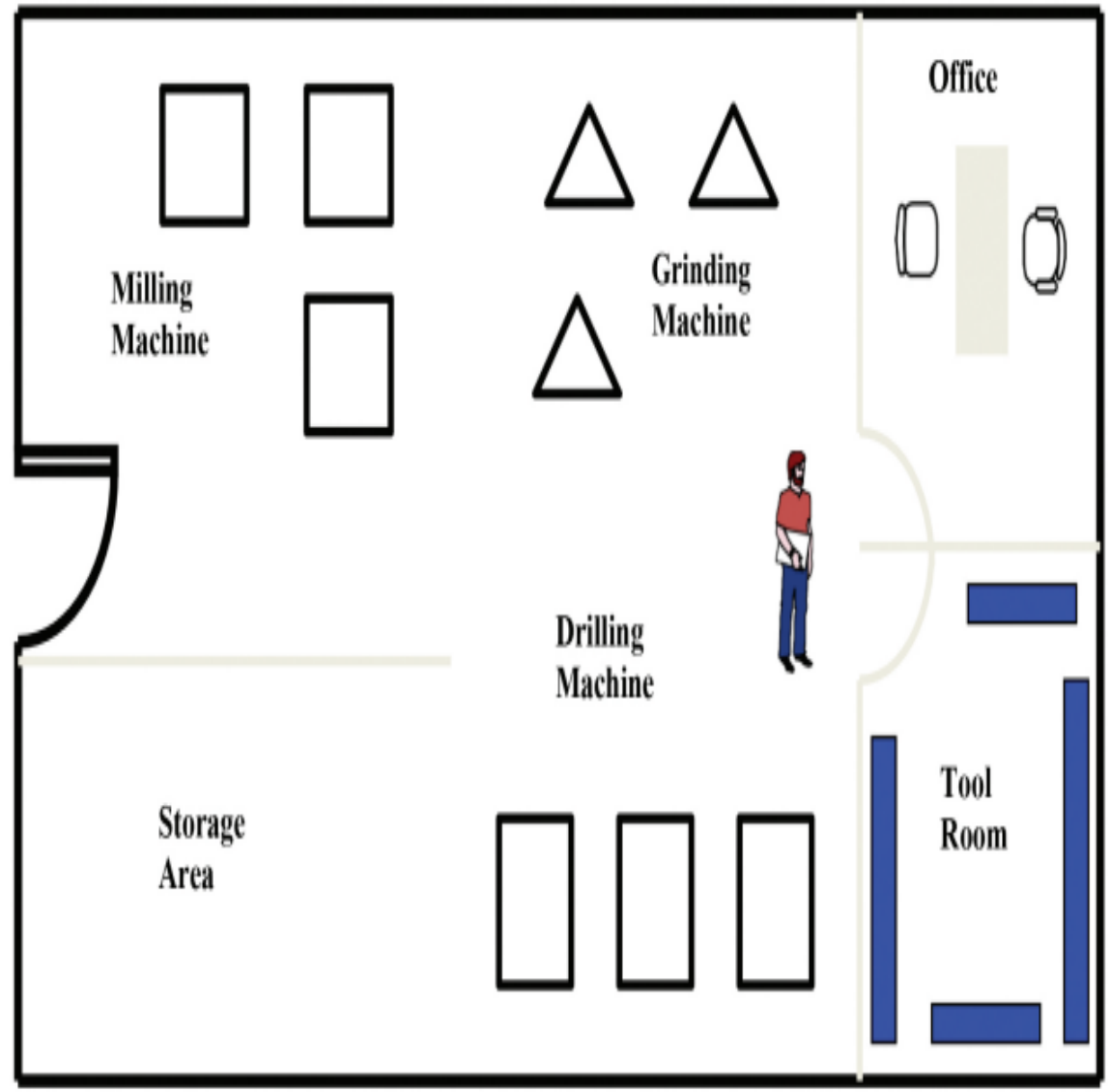

Measure ingredients as accurately as possible and add them in the order specified
Mix the cake batter
Pour the batter evenly into the prepared pans
Bake in the oven for 35 minutes
Check for cake doneness
Let it cool down completely for 15 minutes
Decorate the cake
Place the cake at display area


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2. Operation Layout

- Design of the area for operational activities
- Arrangement of all resources that consume space within a facility - machinery, equipment, workers and
- Huge role in the effectiveness of the operations activities – affects productivity.
- Purpose - use the area efficiently - operation processes will not face wastage in material traveling, delay in operation, excess of time taken for a process and quality deterioration.


Finished
Goods
Storage
Filling Area
Steaming
Office
Fruits Cleaning Area
Receiving
Area
Washing Area
Fruits Chopper


Office
Milling
Machine
Grinding
Machine
Drilling
Machine
Storage
Area
Tool
Room

3. Business & Operation Hours

- Business hours - hours of a business is conducted or the time of the day during which business is transacted.

- Working hours - hours that workers need to work in an organization based on the business hours.



4. License, Permits and Regulations

- Operating a business requires certain licenses, permits and regulations.
- Identify which agency is responsible to approve
- All of the expenses included in the operational budget – preoperational costing
- License/permit/regulations:
  - License for signboard
  - Permits for workers
  - Typhoid injection for the F&B workers

5. Production Planning

- Determine how much output to produce for a certain period of time such as in a day, a week or a month.
- Ensure capable of producing output based on the sales forecast.
- Ultimate goal > produce based on customer's requirement with minimum resources
- Principles of production planning - quantity & timing.

$$\text{Number of outputs per month} = \frac{\text{Sales Forecast per month}}{\text{Price per unit}}$$

$$\text{Output per day} = \frac{\text{Number of outputs per month}}{\text{Working days per month}}$$

$$\text{Output per hour} = \frac{\text{Output produced per day}}{\text{Daily working hour}}$$

Example of calculation for output per day

- **Sales forecast per month = RM 25,200.00**
- **Price per unit = RM 15.00**
- **Number of output per month = RM 25,200.00**

  RM 15.00

  = 1,680 units

- **Working days per month is 24 days**
- **Production output per day : = 1,680 units**

  24 days

  = 70 units per day

6. Manpower Planning

- Placing the right number of people, the right kind of people at the right place, right time.
- Deciding the number of manpower needed to fulfil the production based on production planning - avoid excess of workers and unnecessary costs.

$$\text{No. of Worker Required} = \frac{\text{Output per day}}{\text{Working hours}} \text{ X Worker Production Time / Unit}$$

Planned output per day	= 27 units
Working hours per day	= 8 hours
Units produced per hour	= 4 unit
No. of workers required	= (27 units/8 hours) X (1 hour/4 units)
	= 0.8 (1 worker)

Planned output per day	= 62 cars
Working hours per day	= 10 hours
Units produced per hour	= 8
No. of workers required	= (62 cars/10 hours) X (1 hour/8 cars)
	= 0.77 (1 worker)

7. Machine and Equipment Planning

- Determine number of machines and equipment based on production planning.
- Purchase required number -> no wastage in terms of excess and underutilization of machines and equipment, and also minimizing unnecessary costs

$$\text{No of machine Required} = \frac{\text{Output per day}}{\text{Working Hours} \times \text{Machine Production Time/ Unit}}$$

Planned output per day	=	27 units
Working hours per day	=	8 hours
Units produced per hour	=	4 cloths
Number of sewing machine required	=	(27 units) / (8 hour x 4 units)
	=	0.8 (1 sewing machine)

Planned output per day	=	62 units
Working hours per day	=	10 hours
Units produced per hour	=	7 units
No. of water pressure required	=	(38 units) / (10 hours x 7 units)
	=	0.54 (1 water pressure)

Machine Equipment Consideration:

1. Price
2. Quality and reliability
3. Availability of spare parts
4. Break down maintenance facility
5. Technology & user friendly
6. Supplier reputation
7. After sale services

No.	Machine	Quantity	Price/Unit (RM)	Total (RM)
	Total			

8. Material Planning

- determine the raw materials necessary to produce the desired product.
- Cost of operations is mainly consists of cost of raw materials (60-80%)

Product No	Description	Quantity	Total

9. Operational Budget

Item	Fixed Asset (RM)	Monthly Expenses (RM)	Other Expenses (RM)
Machine and Equipment	XXXX		
Renovation	XXXX		
Raw Materials		XXXX	
Wages+ EPF + SOCSO		XXXX	
Utilities		XXXX	
Rent		XXXX	XXXX
Deposit			
TOTAL AMOUNT	**XXXX**	**XXXX**	**XXXX**

SUMMARY

- Operations play an important role in organization as involve the processes by which people, capital and material are linked to produce goods and services.
- Output should have value added and be worth more than the cost of inputs.
- The ability to produce and deliver high quality & low-cost and meet customer's needs and deliver in a timely manner would have greater probability of achieving its success.



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